Verification Guide
Conditional $\Lambda < 0.116398$

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10 October 2026 — Version 1.0

1. Result and entry points

The main theorem assumes that every nontrivial zeta zero has real part at most 7/8 and gives

$$\Lambda \le 116397726408291/10^{15} = 0.116397726408291 < 0.116398.$$

The alternate ceiling is 0.116397726408312. Each main history has 14,899 steps. No program in this deposit proves the 7/8 hypothesis, repeats the published finite-RH computation, or formalises the complete analytic proof.

Read `Conditional_0116398.pdf` first. The `Matched_Source_Supplement.pdf` and `Prior_Analytic_Foundations.pdf` give the supporting analytic arguments. The latter is an unchanged historical source: its Gaussian majorant and wall-error comparison must be read with Appendix B of the current main paper. The complete current kernel constants are in Appendix A, not just the older catalogue.

`Further_Consequences.pdf` is separate. Its 0.09 implication requires the stronger initial strip $\mathsf{H}_{34/43}$. The proposed all-scale covariance cap is unproved. Neither statement is a premise of the main theorem.

2. Check the deposit without executing mathematical code

Place all upload files together and run:

```
python -B Verify_Deposit.py
```

This standard-library script checks the declared filenames, sizes, SHA-256 values, PDF headers, ZIP integrity and main source/configuration identities. It does not extract or execute scripts inside an archive. Its success establishes integrity, not mathematical correctness. The current manifest and receipt use the new 0.116398 result throughout.

3. Extract and audit the scientific evidence

Extract `Proof_and_Replay.zip`. It contains the single root `BCLD_Certified_Retuning_20261010`. Its contents are byte-identical to the tested research evidence; its internal filenames are retained so that no scientific manifest or certificate needs to be rewritten. Change into that directory. Use ordinary Python with assertions enabled; do not use `-O` or `-OO`. All output directories must be new or empty and outside that root.

```
python -B RUN.py --check-only
python -B RUN.py --audit --output ../retuning_audit
```

The first command checks only internal identities. The second performs these mathematical-data checks: the baseline's complete 13-stage audit, the independent new kernel verification, full retuned-source reconstruction

and selection, the restricted old-table comparison, the covariance constants, and all eight final-source histories.

To regenerate the exact histories as well:

```
python -B RUN.py --audit --regenerate-histories \
  --output ../retuning_full_audit
```

The producer outputs must match all eight supplied compressed histories byte-for-byte. A fresh execution ends with `CERTIFIED_RETUNING_AUDIT_PASS`. The receipt is `../retuning_full_audit/EXECUTION_RECEIPT.json`. Read it together with the stage logs and exact source output, rather than treating the final line alone as proof.

4. Optional fresh transcendental production

The preceding audit consumes stored interval enclosures. To regenerate them at the frozen retuned plan, first finish the full audit above. The native wrapper requires CPython 3.13 on Linux x86-64 and uses the bundled python-flint/Arb/FLINT runtime. It downloads nothing and runs no optimiser. Check compatibility before starting production:

```
python -B code/replay_native.py \
  --audit-dir ../retuning_full_audit \
  --output ../retuning_native --preflight-only
```

Then run:

```
python -B code/replay_native.py \
  --audit-dir ../retuning_full_audit \
  --output ../retuning_native --workers 4
```

The wrapper produces all 608 time cells in each construction in four adjoining blocks, reruns the joint and matched-arithmetic producers, applies the corrected physical-error safety checks, and reconstructs the complete mathematical source. It requires equality with the retained `SOURCE_RETUNED.json`. Receipt times and raw-output hashes may change; those timing fields are not mathematical data. The exact reconstructed source must agree.

The finite-head walls are copied from the retained baseline and checked, not regenerated. No focused-band production is assumed at the changed heights. The original and new source constructions both use Arb/FLINT; two constructions do not mean two independent transcendental libraries. Full native reproduction of the old baseline is a separate optional task in its own retained runner, not a claim made by the new wrapper.

5. Proof-to-evidence map

Paths below are relative to the extracted evidence root.

Obligation	Primary files
Current exact configuration	`evidence/configs/main.json`
Main histories	`evidence/histories/main_primary.json.gz` and `main_alternate.json.gz`
Selected source and provider choice	`evidence/SOURCE_SELECTED.json` and `SELECTION.json`
Retuned plan and raw intervals	`evidence/PLAN_NEW.json` and `RETUNED_NATIVE.zip`
New continuous-domain kernels	`evidence/GEOMETRY_CERTIFICATES.json`; `code/verify_geometry.py`
Source reconstruction	`code/rebuild_source.py`

Obligation	Primary files
Native production wrapper	`code/replay_native.py`
Scalar producer and independent auditor	`baseline/Proof_and_Replay/code/produce_history.py` and `audit_history.py`
Physical analytic sources and runtime	`baseline/Proof_and_Replay/dependencies/Retained_Analytic_Sources.zip`
Stronger-strip histories	`evidence/configs/strip_34_43.json` and matching histories; distinct hypothesis
Covariance and restricted-method checks	`code/check_covariance_bridge.py` and `code/check_table_limit.py`

The other four histories, labelled `strip_63_80` and `strip_79_100`, are additional stronger-strip comparisons. Only `main` proves the current headline from 7/8. Legacy discovery fields named `initial_q` and `strict_bound` inside the source plan are not the current comparison parameters. The current hypothesis, seed and terminal bound are bound by `evidence/configs/main.json` and the complete main histories. The source is used only at its certified time–height cells.

6. Manuscript rebuild and historical labels

Extract `Manuscript_Sources.zip` and run its `BUILD_PDFS.py` with an installed TeX Live environment. It builds the four current PDFs without network access. The standalone main `.tex` includes its own bibliography. The source archive also keeps the unchanged author sources for the two prior PDFs.

This is the first deposit of the 0.116398 paper. No 0.118 paper was published. Older files inside the immutable evidence sometimes use "publication" for an internally prepared package; read those labels as undeposited development history. The current public README, main paper, manifest and citation file take precedence for deposit identity. The September DOI belongs to the separate 0.149396 paper and is never the DOI of this result.

7. Reading the verification receipt

`Verification_Receipt.json` records this deposit preparation's fresh stored-source audit, exact-history regeneration and native-replay outcome, with input identities and execution scope. Historical production receipts remain in the unchanged evidence. Neither receipt is external peer review. Correctness of the stated analytic reductions remains a separate mathematical obligation from execution of these checks.